

Database Management Completed SPARK Reference

YardFresh Community Market - question-by-question completed guide

REFERENCE EXAMPLE ONLY

Use this document to study the wording, steps, structure and quality of completed work. Do not submit, rename or copy this SPARK project as your own SBA.

Project: YardFresh Community Market

Sector: Business

Questions being answered

Task A - Tables

1. Create a database named YardFreshDB_YourFullName.accdb.

How SPARK completed it

- Open your database software and create a blank database.
- Save it immediately with the exact filename in the question.
- Keep the database and all later SBA files inside one organised project folder.

Completed result

SPARK reference filename: YardFreshDB_YourFullName.accdb.

2. Create the Product, Supplier and StockPurchase tables using suitable field names, data types and primary keys.

How SPARK completed it

- Create Product first and enter the fields exactly as planned in your data dictionary.
- Repeat the process for Supplier and StockPurchase.
- Choose the primary key before entering records. Check currency, date and numeric data types carefully.

Completed result

Completed table structures:

Product: ProductID - Short Text, primary key, ProductName - Short Text, Category - Short Text, UnitPrice - Currency, QuantityInStock - Number, ReorderLevel - Number, SupplierID - Short Text, foreign key

Supplier: SupplierID - Short Text, primary key, SupplierName - Short Text, Phone - Short Text, Email - Short Text, Parish - Short Text

StockPurchase: PurchaseID - Short Text, primary key, ProductID - Short Text, foreign key, PurchaseDate - Date/Time, QuantityBought - Number, CostPerUnit - Currency

3. Enter realistic records and apply the validation rules required by the project.

How SPARK completed it

- Enter enough varied records to make your queries meaningful.
- Use validation rules for fields with a limited set or range of allowed values.
- Test the validation by deliberately entering one invalid value and confirming that the software rejects it.

Completed result

The SPARK completed database uses fictional records with different categories, dates, fees and statuses so every query can be tested.

4. Create the required relationships and enforce the correct links between the tables.

How SPARK completed it

- Open the Relationships window.
- Add all tables and drag each primary key to the matching foreign key.
- Check the real-world relationship before selecting referential-integrity options.

Completed result

Supplier.SupplierID 1-to-many Product.SupplierID; Product.ProductID 1-to-many StockPurchase.ProductID.

Task B - Queries and Form

1. Create a query called Low Stock. Show products at or below the reorder level. Include ProductName, Category, QuantityInStock and ReorderLevel.

How SPARK completed it

- Add only the table or tables required by the question.
- Add the requested fields to the query grid.
- Enter the criterion: QuantityInStock <= ReorderLevel.
- Run the query and compare every returned record with the source data.

Completed result

Query name: Low Stock. Criterion used: QuantityInStock <= ReorderLevel.

2. Create a query called Supplier Orders. Join Product, Supplier and StockPurchase. Calculate the order value.

How SPARK completed it

- Add the related tables and confirm the join lines are correct.
- Add the display fields requested in the question.
- Create the calculated field using: OrderValue: [QuantityBought]*[CostPerUnit].
- Run the query and calculate one record manually to verify the result.

Completed result

Query name: Supplier Orders. Calculation used: OrderValue: [QuantityBought]*[CostPerUnit].

3. Create Product Information Form with Supplier subform and clear stock headings.

How SPARK completed it

- Use the main record source for the parent form.
- Use the related table as the subform source.
- Add a clear title, useful instructions and navigation controls.
- Open several records and confirm the subform changes with the main record.

Completed result

Completed form design: Product Information Form with Supplier subform and clear stock headings.

Task C - Report

1. Create Stock Value Report grouped by Category and showing total stock value.

How SPARK completed it

- Use a query as the report source when it already contains the correct fields or calculation.

- Apply the exact grouping and sorting required by the question.
- Add the required summary in the group or report footer.
- Preview the report and correct clipped headings, blank pages and unreadable columns.

Completed result

Completed report: Stock Value Report grouped by Category and showing total stock value.

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Completed database structure

Product

Field	Design
ProductID	Short Text, primary key
ProductName	Short Text
Category	Short Text
UnitPrice	Currency
QuantityInStock	Number
ReorderLevel	Number
SupplierID	Short Text, foreign key

Supplier

Field	Design
SupplierID	Short Text, primary key
SupplierName	Short Text
Phone	Short Text
Email	Short Text
Parish	Short Text

StockPurchase

Field	Design
PurchaseID	Short Text, primary key
ProductID	Short Text, foreign key
PurchaseDate	Date/Time
QuantityBought	Number
CostPerUnit	Currency

Relationships

Supplier.SupplierID 1-to-many Product.SupplierID; Product.ProductID 1-to-many StockPurchase.ProductID.

Sample records

ProductID	ProductName	Category	UnitPrice	QuantityInStock	ReorderLevel
PR001	Rice 2kg	Grocery	980	25	10
PR002	Plantain Chips	Snacks	350	8	12
PR003	Fresh Juice	Beverage	500	18	8
PR004	Ground Coffee	Beverage	1800	6	8
PR005	Bath Soap	Household	420	30	10
PR006	Cereal	Grocery	1200	9	10

Completed queries

Low Stock

Show products at or below the reorder level. Include ProductName, Category, QuantityInStock and ReorderLevel.

Criterion: QuantityInStock <= ReorderLevel

Supplier Orders

Join Product, Supplier and StockPurchase. Calculate the order value.

Calculation: OrderValue: [QuantityBought]*[CostPerUnit]

Completed form

Product Information Form with Supplier subform and clear stock headings.

Completed report

Stock Value Report grouped by Category and showing total stock value.

Final section check

- Every question in the practice brief has a visible completed result.
- The output uses the same project facts as the other sections.
- The work has been tested or checked rather than assumed correct.
- This SPARK example is watermarked and is for study only. Build your own teacher-assigned SBA.

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